Ties and quantization in the ordinal complexity–entropy plane: exact finite-sample identities and a discretization correction

D. Sierra-Porta[1,*], A. M. S. Borin Jr.[2], and Other?????.[2]

[1]Universidad Tecnológica de Bolívar, Cartagena de Indias, 130010, Colombia
[2]Federal Institute of Triángulo Mineiro (IFTM), Uberaba, MG 38064-790, Brazil
[*]Corresponding author: dporta@utb.edu.co

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Abstract

Ordinal (Bandt–Pompe) analysis presumes a series without ties, yet the records it is most often applied to are quantized: interbeat intervals in integer samples, photon or particle counts, coarsely digitized signals. Quantization produces ties, and the common remedy of breaking them by an arbitrary rule displaces the point a series occupies in the complexity–entropy plane. We treat ties as part of the alphabet rather than as noise, working with generalized ordinal patterns (weak orders), and give the finite-sample theory of the plane on that alphabet. The exact bias of the coordinates under overlapping windows carries over from the permutation case, with lagged collision terms that we compute in closed form for quantized continuous sources. The larger effect is separate from that overlap: discretization moves the two coordinates on different scales, the disequilibrium linearly in the quantization step and the normalized entropy like $\Delta \log(1/\Delta)$, with universal coefficients that depend on the source only through the collision integral of its density, and the linear part is correctable back to the continuous plane. Ties also invalidate the calibrated null used for continuous series, and a tie-matched permutation null restores correct size while keeping power. We validate the identities against ground-truth processes and apply them to interbeat-interval records from two heart-rate databases at different sampling lattices, where the tie-aware plane is single-valued and the conventional tie-broken plane is not.

1 Introduction

The complexity–entropy plane pairs a normalized permutation entropy with a distance to the uniform ordinal distribution, and it has become a standard chart for separating structured from unstructured dynamics [1–5], with applications that run from electroencephalography and heart-rate variability to finance and the geosciences [6–8]. Its construction assumes that the series has no ties: the ordinal pattern of a window is the permutation that sorts it, and a permutation exists only when all values differ. Many of the records the plane is applied to do not meet this assumption. Interbeat intervals are reported in integer samples [9, 10], particle and photon counts are integers by construction, and digitized signals live on a fixed grid. In each case the values fall on a lattice, equal values occur, and the ordinal pattern is not defined until the ties are resolved.

The usual resolution is to break ties by an external rule, most often the order in which the equal values appear. This is not a neutral choice. Breaking ties by position imposes an ordering that the data do not contain, and the imposed order lowers the estimated entropy and raises the estimated disequilibrium, moving the series to a different point of the plane; the size of the move grows with the fraction of tied windows [11]. Two lines of work respond to this. One keeps the permutation alphabet and studies how tie policies and finite samples distort the coordinates [11–13]. The other enlarges the alphabet so that a tie is its own symbol: a window maps to a weak order rather than a permutation, and the pattern probabilities and their asymptotic dependence structure have been developed for this generalized construction [14–16].

Neither line yields what an analysis of a quantized short record requires. The finite-sample theory of the plane, including the exact bias that overlapping windows introduce, has been worked out for the permutation alphabet [17] but not for the alphabet that admits ties. The generalized alphabet has been given an asymptotic dependence theory [14], but not a finite-sample account of the plane coordinates, and neither line asks how the quantization step itself relocates a series in the plane. The step is the controllable quantity in a digitized experiment, and its effect is separate from the dependence between overlapping windows.

This paper supplies both pieces on the tie-admitting alphabet, and its central claim is that discretization, not window overlap, is what relocates a short series in the ordinal plane, with an exact theory and a first-order correction for it. We (i) set the plane on the generalized-ordinal-pattern alphabet and identify the continuous-limit reference point; (ii) give the exact finite-sample bias of the coordinates under overlapping windows, with collision terms computed in closed form for quantized continuous sources; (iii) show that discretization moves the two coordinates on different scales, the disequilibrium linearly in the step and the entropy like $\Delta \log(1/\Delta)$, with universal coefficients set only by the collision integral of the source density, and read off the correction to the continuous plane; (iv) show that the calibrated null used for continuous series is invalid once ties are present and replace it with a tie-matched permutation null; and (v) validate the identities against ground-truth processes and apply them to a cohort of interbeat-interval records across two databases.

2 The generalized-ordinal-pattern plane

Fix an embedding dimension $m \geq 2$ and read the series at consecutive starting points and unit delay, so that window t is $(x_t, \ldots, x_{t+m-1})$. When the values are distinct the window has a permutation pattern; when some coincide it has a weak order, obtained by assigning each position the rank of its value among the distinct values present, with equal values sharing a rank. The set of weak orders of m elements is the alphabet, and its size is the ordered Bell (Fubini) number $M = a(m)$, which is $1, 3, 13, 75, 541$ for $m = 1, \ldots, 5$, against $m! = 1, 2, 6, 24, 120$ for the strict orders. Ties therefore enlarge the alphabet, from 24 to 75 patterns at $m = 4$, and the extra patterns are exactly those a tie-breaking rule would suppress.

Let p_π be the stationary probability of pattern π and $\hat{p}_\pi = c_\pi/n$ its relative frequency over the $n = N - (m-1)$ windows of a length-N record. The plane coordinates are the normalized permutation entropy, the disequilibrium, and the Jensen–Shannon statistical complexity,

$$H = \frac{-\sum_\pi p_\pi \log p_\pi}{\log M}, \qquad D = \sum_\pi \left(p_\pi - \tfrac{1}{M}\right)^2 = \sum_\pi p_\pi^2 - \tfrac{1}{M}, \qquad C_{\mathrm{JS}} = Q_0 \, H \, \mathcal{J}(p), \tag{1}$$

where $\mathcal{J}(p)$ is the Jensen–Shannon divergence between p and the uniform distribution on the M patterns and Q_0 its normalizing constant [4], both evaluated on the enlarged alphabet so that $\log M = \log a(m)$ replaces $\log m!$. The normalization matters: the uniform reference is now uniformity over all $a(m)$ weak orders, a distribution a tie-free series never approaches.

A continuous source has no ties, so it realizes only the $m!$ strict orders and leaves the remaining $a(m) - m!$ tie patterns empty. Independent continuous white noise makes the strict orders equiprobable, which fixes the reference point that quantization will move,

$$H_0 = \frac{\log m!}{\log a(m)}, \qquad D_0 = \frac{1}{m!} - \frac{1}{a(m)}. \tag{2}$$

For $m = 4$ this is $H_0 = 0.7361$ and $D_0 = 0.02833$. The value $H_0 < 1$ records a structural fact rather than a defect: on the tie alphabet the most disordered continuous process is not at maximum entropy, because half of its patterns cannot occur. Section 4 shows how quantization fills those patterns and where it takes the coordinates.

3 Finite-sample bias under overlapping windows

Overlapping windows share data, so their patterns are dependent and the plug-in coordinates are biased at finite N. The variance of the permutation-entropy estimate and its dependence on the ordinal pattern set have

been quantified for the permutation alphabet [18], and the exact finite-sample bias of the plane coordinates under overlap has a closed form in terms of lagged pattern-collision probabilities [17]. The derivation uses only the alphabet through the collision probabilities, so it transfers unchanged to the weak-order alphabet. With $s_2 = \sum_\pi p_\pi^2$ and the lagged collision term $g(h) = \Pr(\Pi_t = \Pi_{t+h}) - s_2$, the plug-in and distinct-pairs estimators of the disequilibrium satisfy

$$E\{\hat{D}_{\text{plug}}\} - D = \frac{1 - s_2}{n} + \frac{2}{n^2} \sum_{h=1}^{m-1} (n - h)\, g(h), \qquad E\{\hat{D}_U\} - D = \frac{2}{n(n-1)} \sum_{h=1}^{m-1} (n - h)\, g(h), \tag{3}$$

for an independent source, where the sums run only to $m-1$ because non-overlapping windows do not collide. The gap estimator that excludes lags up to $w = m - 1$ is unbiased for such a source. What the tie alphabet changes is the content of s_2 and $g(h)$, not the form of (3).

For a source that is a continuous process quantized to a step Δ, both quantities are finite sums that we evaluate exactly. Each value falls on lattice level k with probability $q_k = F((k + \frac{1}{2})\Delta) - F((k - \frac{1}{2})\Delta)$, F the source distribution. Because the values are independent, the probability that u of them realize a weak order with ordered tie-group sizes $(b_0, \ldots, b_{g-1})$ is

$$P(b) = \sum_{k_0 < k_1 < \cdots < k_{g-1}} \prod_{j=0}^{g-1} q_{k_j}^{b_j}, \tag{4}$$

a sum over strictly increasing levels that a single pass over the lattice evaluates. Summing $P(\cdot)^2$ over the $a(m)$ weak orders of a window gives s_2, and summing $P(\cdot)$ over the weak orders of the $(m + h)$-value union of two windows separated by h, restricted to those in which the two sub-windows carry the same pattern, gives $\Pr(\Pi_t = \Pi_{t+h})$ and hence $g(h)$. As $\Delta \to 0$ the tie patterns lose their probability, and $s_2 \to 1/m!$ with $g(h)$ returning the continuous permutation values, so (3) reduces to the tie-free case.

Table 1 compares the exact biases from (3) with direct simulation for Gaussian white noise at $m = 3$ and $m = 4$. The two agree within the Monte Carlo error at every step, which confirms the closed form. The distinct-pairs bias changes sign as ties become frequent, negative at small step and positive once ties dominate, so the estimator that is unbiased for a continuous independent source acquires under quantization a bias whose direction depends on the step.

Table 1: Exact finite-sample bias of the disequilibrium on the tie alphabet against Monte Carlo, for Δ-quantized Gaussian white noise ($N = 200$). The plug-in and distinct-pairs biases keep the permutation-case form; only s_2 and the collision terms $g(h)$ change. As $\Delta \to 0$ the terms recover the continuous permutation values and $s_2 \to 1/m!$.

		plug-in bias (10^{-3})		U **bias** (10^{-3})	
m	Δ	**exact**	**simulation**	**exact**	**simulation**
3	0.5	+3.796	+3.781	−0.822	−0.837
3	0.75	+3.927	+3.928	−0.734	−0.734
3	1.0	+3.993	+3.980	−0.655	−0.668
3	1.5	+4.307	+4.266	−0.216	−0.258
4	0.5	+4.785	+4.780	−0.221	−0.226
4	0.75	+4.791	+4.779	−0.212	−0.225
4	1.0	+4.793	+4.786	−0.191	−0.199
4	1.5	+5.061	+5.053	+0.160	+0.152

4 The discretization law

Section 3 treated dependence between windows. We now hold the source independent and ask a separate question: as the quantization step grows from zero, where do the coordinates go? The answer is governed by

a single small quantity, the probability that two independent values coincide,

$$c(\Delta) = \sum_k q_k^2 = \Delta \int f(x)^2 \, dx + O(\Delta^2), \tag{5}$$

the second form being the Riemann sum of the squared density $f = F'$. The collision integral $\int f^2$ is the only property of the source that enters at leading order; it is $1/(2\sqrt{\pi})$ for a unit Gaussian, $\sqrt{2}/4$ for a unit Laplace, $1/(2\sqrt{3})$ for a unit uniform. The expansions below require a square-integrable and bounded density, $f \in L^2 \cap L^\infty$, so that $\int f^2$ is finite and the higher-order remainders are uniform in the pattern; they degrade for heavy-tailed or near-discrete sources, where c is not small.

At order Δ a window has at most one tied pair, since two coincidences cost $O(\Delta^2)$. A single tie takes a strict order to a one-tie weak order, so probability leaks out of the $m!$ strict patterns into the tie patterns at total rate $\binom{m}{2}c$, distributed by symmetry equally among the strict orders. Each strict pattern therefore keeps $1/m! - \binom{m}{2}c/m!$, and the tie patterns each carry $O(c)$. These pattern masses to first order in c follow from the exchangeability of the independent sample, proved in Lemma 1 of Appendix A, where the $O(c^2)$ remainders are shown uniform in the pattern. The disequilibrium is quadratic in the probabilities, so the tie patterns contribute at $O(c^2)$ and only the depleted strict patterns matter at first order,

$$D(\Delta) = D_0 - \frac{c(\Delta)}{(m-2)!} + O(c^2). \tag{6}$$

The shift is linear in the step (Proposition 1, Appendix A). The linearity is worth marking against the classical case: Sheppard's correction for moments under grouping is of order Δ^2 [19], because it comes from the variance added within a bin, whereas the ordinal effect is of order Δ, because it comes from coincidences across bins.

Read backwards, (6) recovers the continuous-plane disequilibrium from a quantized record. With n_k the counts of the N raw samples on lattice level k, the unbiased U-statistic for the coincidence probability is $\widehat{c} = \sum_k n_k(n_k-1)/[N(N-1)]$, the same collision device that defines $\widehat{D}_U$ in Sec. 3, and the corrected estimator is

$$\widehat{D}_0 = \widehat{D}_{\mathrm{gap}}(\Delta) + \frac{\widehat{c}}{(m-2)!}, \tag{7}$$

where $\widehat{D}_{\mathrm{gap}}$ is the gap estimator with $w = m-1$, which removes the overlap bias and estimates the population $D(\Delta)$ since quantized independent values are independent beyond the embedding span. Its bias is the $O(c^2)$ residual the linear correction leaves, and the added variance is $O(1/N)$ with an $O(c^2)$ coefficient, negligible against the gap term. Monte Carlo on quantized Gaussian white noise ($m = 4$, $N = 4000$) confirms both the gain and its range: the correction removes 92% of the shift at $\Delta = 0.04$ (tie fraction 7%) and 83% at $\Delta = 0.08$ (13%), and the residual grows like Δ^2, so by $\Delta = 0.16$ (25% ties) only 63% is removed and the linear term begins to overshoot. The correction is therefore a small-step device, dependable up to a tie fraction of roughly a quarter; beyond that the honest move is to report the tie fraction and use the tie-matched null of Sec. 5 rather than to correct.

The entropy behaves differently. The patterns that quantization activates each enter with probability of order c and contribute $-p\log p$, and $\sum(-p\log p)$ over them is of order $c\log(1/c)$ rather than c. Collecting the first two terms (Proposition 2, Appendix A),

$$\big(H(\Delta) - H_0\big)\log M = \binom{m}{2}\Big[\, c(\Delta)\log\tfrac{1}{c(\Delta)} + \big(1 - \log m\big)\, c(\Delta)\Big] + O(c^2). \tag{8}$$

The entropy rises faster than the disequilibrium falls, as $\Delta\log(1/\Delta)$ against Δ, so near the continuous point the plane trajectory leaves along the entropy axis. The coefficients $1/(m-2)!$ and $\binom{m}{2}$ in (6) and (8) are combinatorial and do not depend on the source; the source enters only through $c(\Delta)$, that is, only through $\int f^2$. The second term in (8) is what the leading behavior alone omits. Since $1 - \log m < 0$ for $m \geq 3$, the finite-step ratio $(H-H_0)\log M/[\binom{m}{2}c\log(1/c)] = 1+(1-\log m)/\log(1/c)+O(c/\log(1/c))$ approaches its limit only as $O(1/\log(1/c))$, from below: the entropy coefficient is reached slowly, which is the slow convergence

visible in Fig. 1(b). This is the reason the disequilibrium, whose shift is linear and whose correction is robust, is the coordinate to correct in practice, while no entropy analogue is recommended at usable record lengths.

Figure 1 tests both laws. Panel (a) overlays the exact coordinates and the predictions for Gaussian white noise; for the entropy the two-term law of (8) tracks the exact shift across the small-step regime, where the leading term alone already departs. Panel (b) divides out c and shows the normalized coefficients approaching $1/(m-2)!$ and $\binom{m}{2}$ for Gaussian, Laplace and uniform sources, confirming that the density enters only through the collision integral. The entropy coefficient approaches its limit from below along the predicted $1+(1-\log m)/\log(1/c)$ curve, the slow $O(1/\log(1/c))$ convergence of (8), while the disequilibrium coefficient sits at $1/(m-2)!$ already at the smallest steps. Figure 2 follows the exact trajectory across the full range of Δ. It is not monotone: the entropy rises to a maximum near a tie fraction of 60% and then falls as the coarsening concentrates the distribution on a few patterns. The point (H_0, D_0) is thus not an endpoint but an interior origin from which quantization first increases the apparent entropy, a behavior that a tie-breaking analysis, which never leaves the permutation alphabet, cannot represent.

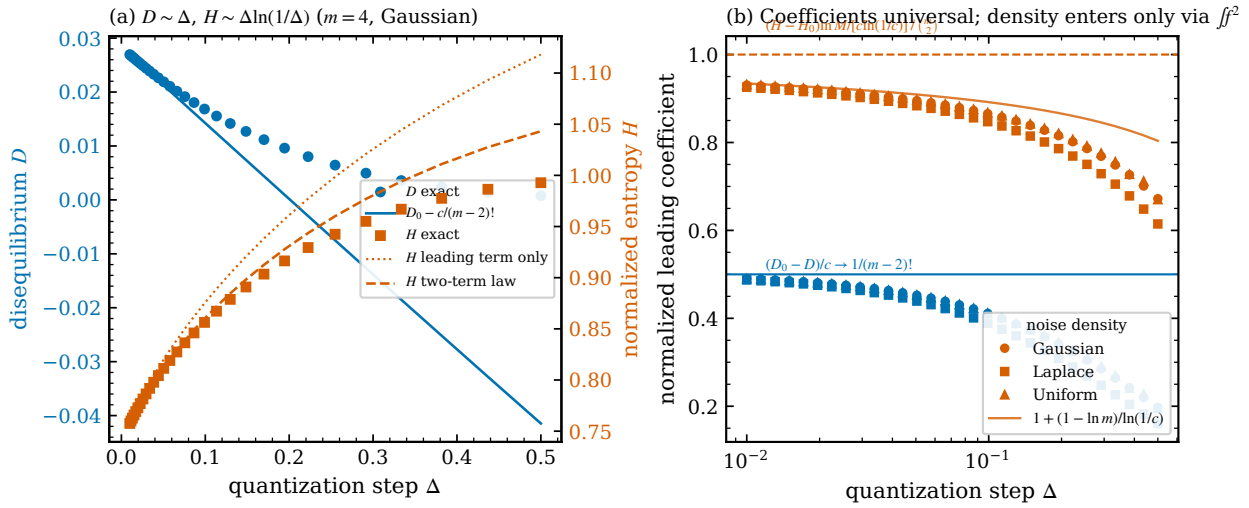


Figure 1: The discretization law. (a) For Gaussian white noise the disequilibrium D follows the linear law $D_0 - c/(m-2)!$ while the entropy follows the two-term law of (8) (dashed), which tracks the exact shift where the leading term alone (dotted) already departs; $c(\Delta) = \sum_k q_k^2$ is the per-pair coincidence probability. (b) Normalizing by c, the disequilibrium coefficient sits at $1/(m-2)!$ while the entropy coefficient approaches $\binom{m}{2}$ from below along $1 + (1 - \log m)/\log(1/c)$ (solid), the slow $O(1/\log(1/c))$ convergence; for Gaussian, Laplace and uniform sources alike, so the coefficients are universal and the density enters only through $\int f^2$.

5 Calibrated inference: the tie-matched null

A point in the plane is read against a null. For continuous series the reference is independent white noise, and a one-sided test on the score $T = (1-H)^2 + C_{\mathrm{JS}}^2$ flags a series that sits far from uniformity [17]. Section 4 makes this reference wrong for quantized data: quantization moves the null itself, so a genuinely independent but quantized series is not near the continuous-white-noise cloud. The consequence is not conservatism. Testing quantized white noise against the continuous null, the observed score lands in the opposite tail, because ties raise H and shrink T; a two-sided reading then rejects the series as anomalously uniform. The null cannot separate the marginal, which carries the ties, from the serial structure it is meant to detect.

The remedy is to match the null to the marginal, in the spirit of surrogate-data testing [20]. Shuffling the observed values gives a surrogate with the same multiset, hence the same ties, and independent order; the permutation null built from many such shuffles asks whether the ordering carries structure beyond what the marginal already implies. Figure 3 reports size and power at $N = 300$. Under quantized white noise,

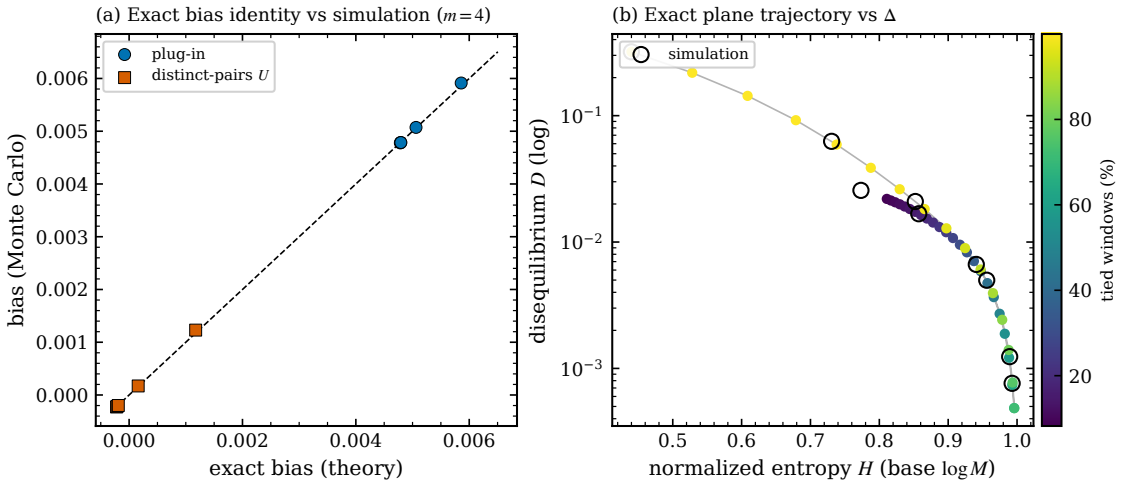


Figure 2: The exact plane trajectory as the quantization step grows. The tie-aware coordinates trace a non-monotonic path: the maximally random continuous process is not at maximum entropy on the tie alphabet, and coarser quantization first raises the apparent entropy by activating tie patterns. Open circles: direct simulation; filled points: the exact theory of Sec. 4.

where the serial null is true, the permutation test holds its nominal level across tie fractions from zero to 98% (rejection 0.048 to 0.062 at $\alpha = 0.05$), while the continuous null does not: one-sided it never rejects, two-sided it rejects every series. Against quantized AR(1) the permutation test keeps its power, 0.74 at $\phi = 0.3$ and above 0.99 for $\phi \geq 0.5$. On quantized data the defensible statement about the plane is therefore a permutation test, not a comparison to a continuous null.

6 Ground truth and a real application

6.1 Ground-truth processes

Table 2 lists the tie-alphabet coordinates of representative processes. The Gaussian rows, white noise and AR(1), are exact: a length-m window of a stationary AR(1) process is multivariate normal with correlation $\phi^{|i-j|}$, and after quantization each pattern probability is a sum of multivariate-normal box probabilities over the lattice-level tuples that induce it, which the Markov structure evaluates through a transfer operator [21, 22].[1] The Poisson-count and logistic-map rows are by long simulation, since their joint lattice law is not multivariate normal. Where both methods apply they agree to the fourth decimal, so the simulation is trustworthy at the same precision for the rows that have no closed form.

The processes occupy distinct regions, and the pattern of occupancy is itself informative. Quantized white noise and Poisson counts sit near the reference point, at high entropy and small disequilibrium: their ordinal patterns stay close to uniform over the realizable orders, and the ties they carry spread probability without concentrating it. The AR(1) processes move toward lower entropy and larger disequilibrium as the correlation or the step grows, tracing the persistence of the process. The logistic map, being deterministic, visits few of the available patterns and sits at low entropy and large disequilibrium. A reading that recorded only the tie fraction would miss this: Poisson counts at $\lambda = 5$ are tied in 59% of windows yet sit close to the noise floor, while the logistic map is tied in only 10% yet is among the most structured processes in the table.

The overlap-bias estimators of Section 3 behave on AR(1) as their derivation predicts only while the dependence stays within the window. At $\phi = 0.5$ the gap estimator with $w = m-1$ recovers the true disequilibrium of a short record, because the correlation has decayed within the embedding span. At $\phi = 0.9$ the correlation

1. Each lattice bin is integrated with a fixed q-node Gauss–Legendre rule; the coordinates are converged in q, changing below the fourth decimal between $q = 6, 8, 12$, so $q = 8$ is used throughout.

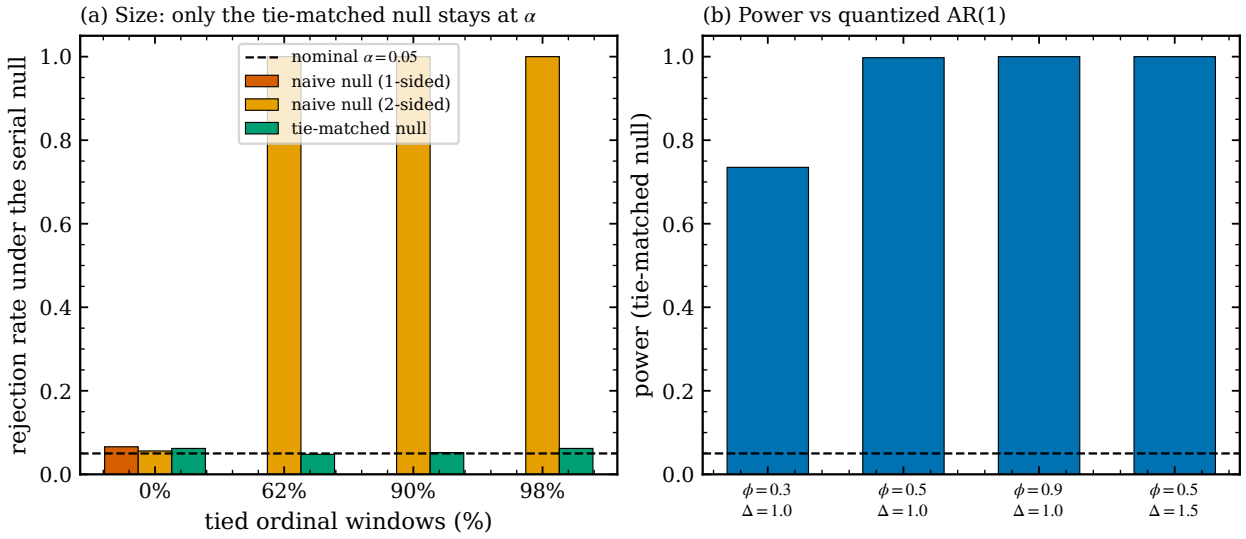


Figure 3: Calibrated inference on quantized data. (a) Under a true IID (but quantized) process the tie-matched permutation null holds size at the nominal $\alpha = 0.05$ across tie fractions, while the tie-free null does not: two-sided it rejects every series. (b) Power of the tie-matched null against quantized AR(1).

reaches beyond the excluded lags, and on records of $N = 300$ the gap leaves a residual of about -1.5×10^{-3} at $\Delta = 0.5$ and -8.5×10^{-3} at $\Delta = 1.0$, growing with the step. No estimator is automatically exact for a process whose dependence exceeds the window, so the calibrated null of Section 5, not any single estimator, is the statement that holds.

6.2 Application: interbeat intervals

Interbeat (RR) intervals are a direct test, because they are recorded in integer samples and carry genuine cardiac dynamics, and ordinal methods have been applied to them before [9, 10]. We take RR series from two open heart-rate databases that differ in their sampling lattice: the Normal Sinus Rhythm RR Interval Database [23] (54 records, ECG at 128 Hz, a 7.8125 ms lattice) and the Fantasia database [24, 25] (7 records, ECG at 250 Hz, a 4 ms lattice), both from PhysioNet [26]. Each series is reduced to normal-to-normal intervals, with a physiological range filter and a relative-jump filter removing ectopic and artifactual beats; we drop the offending intervals rather than replace them, since imputing a central value would manufacture the ties the analysis is about. On one record (tied in 56%) the tie-aware coordinates are $H = 0.904$, $D = 0.022$ and $C_{\mathrm{JS}} = 0.112$, and the plug-in, distinct-pairs and gap disequilibria agree to the third decimal, so the overlap correction is negligible: when the structure is strong the finite-sample overlap bias is small against the signal, and what decides the reading is how the ties are handled.

Two features of the cohort bear on the theory. First, the two databases occupy different tie bands, the Normal Sinus Rhythm records at 41–82% of windows tied (median 65%) and Fantasia at 28–56% (median 45%), and the ordering is the one the discretization law predicts: the coarser 7.8 ms lattice coincides more often than the finer 4 ms one, so the tie fraction is set by the quantization step for a structural reason and not by the physiology alone. Second, the conventional analysis, which breaks ties to remain on the permutation alphabet, does not return a single answer. Resolving ties by order of appearance rather than by added jitter changes the entropy, and the signed difference between the two policies grows with the tie fraction, correlating -0.87 across the 61 records (Fig. 4). The tie-aware treatment removes this ambiguity because it never imposes an order the data do not contain.

Against the tie-matched null the observed series is distinguishable from its surrogate, read two-sided, in 59 of the 61 records, so the ordinal structure is real and not an artifact of the ties. The direction is itself informative: in 58 of them the series sits at lower score than its surrogate, at higher entropy and smaller complexity, the signature of the beat-to-beat regulation of sinus rhythm, whose intervals alternate more than

independent sampling would. The naive one-sided reading, which asks only whether a series is anomalously far from uniformity, would miss this majority; once ties are present the calibrated null must be read on both sides, as Sec. 5 requires.

Table 2: Tie-alphabet coordinates ($m = 4$) for representative processes. The Gaussian rows (white noise and AR(1)) are exact, from multivariate-normal box probabilities on the lattice; the Poisson and logistic rows are by long simulation. Where both apply the two methods agree.

process	**tied windows (%)**	H	D
white noise, $\Delta = 0.5$	62.2	0.9928	7.65×10^{-4}
white noise, $\Delta = 1.0$	89.7	0.9560	4.99×10^{-3}
AR(1) $\phi = 0.5$, $\Delta = 0.5$	71.9	0.9799	2.39×10^{-3}
AR(1) $\phi = 0.9$, $\Delta = 0.5$	94.6	0.8516	1.89×10^{-2}
AR(1) $\phi = 0.9$, $\Delta = 1.0$	99.9	0.6137	1.04×10^{-1}
Poisson counts, $\lambda = 5$	58.5	0.9886	1.20×10^{-3}
Poisson counts, $\lambda = 10$	44.7	0.9633	4.00×10^{-3}
logistic map, $\Delta = 0.02$	10.4	0.6317	7.05×10^{-2}
logistic map, $\Delta = 0.05$	21.0	0.6933	5.36×10^{-2}

7 Discussion

Two effects displace a short quantized series in the ordinal plane, and the paper has kept them apart. Overlapping windows make the plug-in coordinates biased, and the bias keeps the closed form it has on the permutation alphabet, with collision terms we evaluate exactly for a quantized continuous source. Quantization displaces the coordinates on its own, independently of the overlap, and this is the effect a tie-breaking analysis cannot represent, because it stays on the permutation alphabet where the tie patterns do not exist. The quantization shift is linear in the step for the disequilibrium and of order $\Delta \log(1/\Delta)$ for the entropy, with coefficients fixed by the embedding dimension and a source that enters only through its collision integral, and the linear part is correctable. Both shifts follow from a single exchangeability argument (Lemma 1, Appendix A): the linear disequilibrium law of Proposition 1 and the two-term entropy law of Proposition 2, whose second term sets how slowly the entropy coefficient is reached and is why the correction is carried only for the disequilibrium.

These results connect three lines that had not met. The study of tie policies on the permutation alphabet [11–13] established that breaking ties moves the plane and that the choice must be reported, but it remained on the alphabet the ties break; here the ties define the alphabet, and the policy question does not arise. The generalized-ordinal-pattern line [14–16] built the tie-admitting alphabet and its asymptotic dependence theory; we add the finite-sample behavior of the plane coordinates on that alphabet and their dependence on the quantization step. The finite-sample treatment of the plane under overlap [17] supplied the bias machinery for permutations, which we carry over and then set beside the discretization law that is specific to discreteness.

The discretization effect is also distinct from the undersampling that the entropy-estimation literature corrects. Finite samples bias the plug-in entropy downward because rare categories go unseen, and corrections for this are well developed [18, 27, 28]. That bias vanishes as N grows at fixed alphabet. The displacement studied here does not: it is a property of the quantization step, present in the population coordinates themselves, and it survives in the limit of infinite data. Both act on a short quantized record at once, the overlap and undersampling biases shrinking with N while the quantization displacement stays, and only the last is removed by the correction of Section 4.

For practice the implication is a short checklist. An ordinal analysis of a quantized record should state the embedding and the delay, which set what the window resolves [29, 30]; the alphabet, since the tie-aware and tie-broken alphabets are different objects; the tie fraction, and the policy if ties are broken; the estimator;

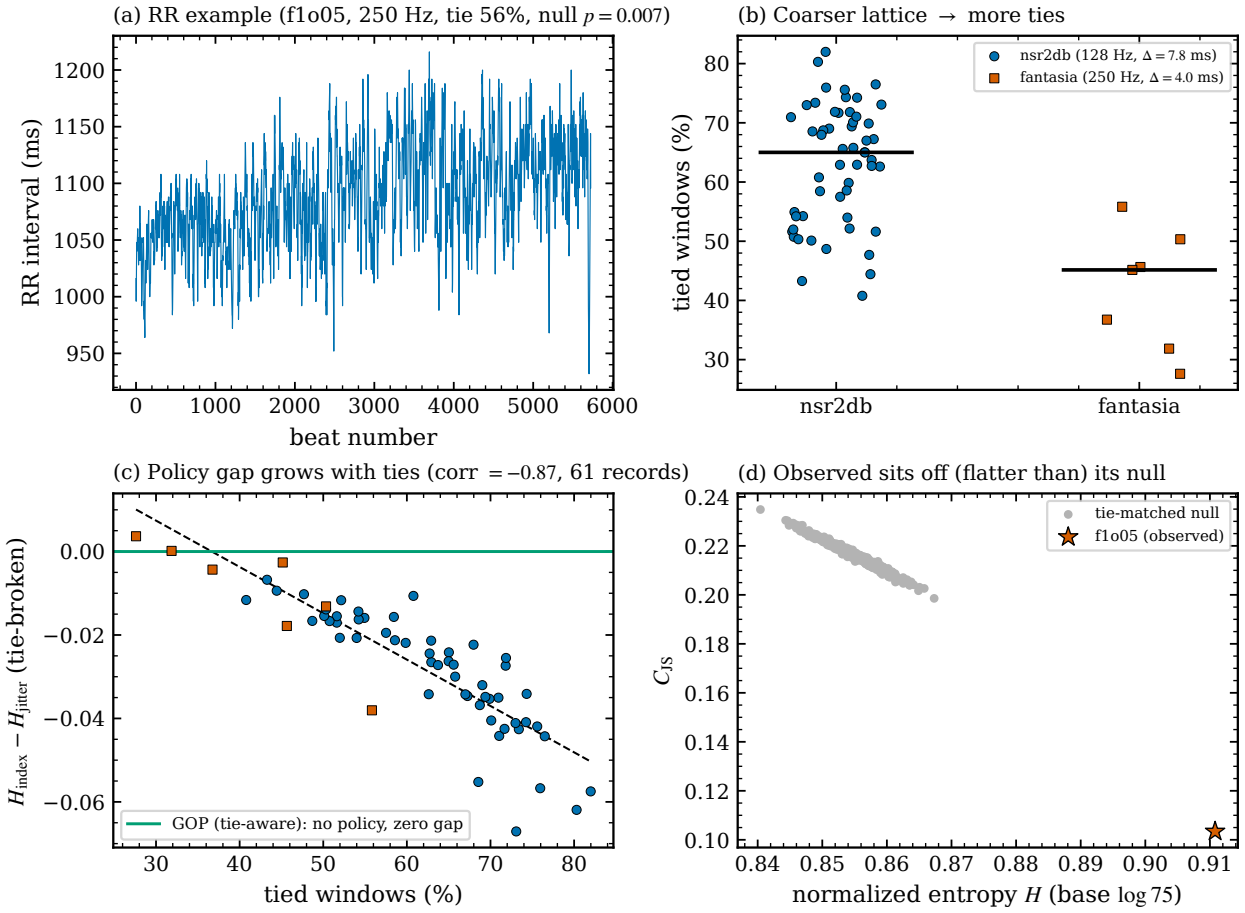


Figure 4: Interbeat-interval cohort from two PhysioNet databases at different lattices (Normal Sinus Rhythm, 128 Hz; Fantasia, 250 Hz). (a) An example record. (b) The databases occupy different tie bands, the coarser lattice tied more often, as the discretization law predicts. (c) The conventional tie-broken plane is policy-dependent: across the 61 records the signed index−jitter entropy gap grows with the tie fraction (correlation −0.87), while the tie-aware treatment needs no policy (zero gap). (d) The example sits off its own tie-matched null, on the flatter side; read two-sided the null separates the series from its surrogate in 59 of 61 records.

and the quantization step together with the correction it implies. Reporting the tie fraction by itself, as is sometimes advised, does not place the result, because the same tie fraction is compatible with very different points depending on the process, as Table 2 shows. Inference should be made against a tie-matched null. These are light requirements, and they matter most where the plane is most used, in the short, quantized biomedical and econophysics series that motivate much of the field [6–8].

The closed-form collision terms assume a source that is a continuous process read on a lattice, and cover any independent quantized process through its marginal. A dependent process needs its joint law on the lattice, which is available when that law is Gaussian, as the exact AR(1) coordinates show; a dependent process with a non-Gaussian joint law, such as an integer-valued count model, still requires simulation for the coordinates, with the permutation null for inference. The entropy law carries a logarithm and, by Proposition 2, approaches its leading coefficient only as $O(1/\log(1/c))$, so the correction is most reliable for the disequilibrium, and on very short records the estimates are further limited by patterns that are never visited. The analysis is for unit embedding delay, where quantization and ties are the whole story. How the tie alphabet interacts with a longer delay, which sets the timescale a window resolves and is the lever that exposes structure at the native cadence of a slow process, is the natural continuation.

8 Conclusions

Treating ties as part of the ordinal alphabet, rather than as noise to be broken, turns the quantization of a short series from an ambiguity into a quantity with an exact theory. On the generalized-ordinal-pattern alphabet the finite-sample overlap bias of the plane coordinates keeps its closed form, with collision terms we compute exactly for quantized continuous sources. Separately, and this is the main point, the quantization step relocates the coordinates on its own: it moves the disequilibrium linearly and the entropy like $\Delta \log(1/\Delta)$, with coefficients that depend on the source only through its collision integral, and the linear shift is correctable. The null used for continuous series fails once ties are present, and a tie-matched permutation null restores valid inference with correct size and retained power. Across interbeat-interval records from two databases the tie-aware plane is single-valued while the conventional tie-broken plane depends on an arbitrary policy, the more so as ties grow, and the tie-matched null read two-sided separates the series from its surrogate. The cost of the tie-aware treatment is a larger alphabet and the reporting of a few quantities; the return is that a quantized short series has a definite, reproducible place in the plane.

Two consequences follow for how the plane is used. Because the overlap bias and the discretization shift have different signatures, one shrinking with the record length and the other fixed by the step, they call for different remedies and should not be conflated: the gap estimator addresses the first and the collision correction the second, and reporting the quantization step is what lets a reader tell which is operating. Because the correction returns the continuous-plane disequilibrium, coordinates measured on instruments that digitize at different resolutions become comparable on a common reference, which a tie-breaking analysis cannot provide. The account given here is for unit embedding delay and for sources whose joint lattice law is independent or Gaussian; a longer delay, which resolves structure at a slower cadence, and dependent sources with a non-Gaussian joint law, whose coordinates at present need simulation, are where the tie-aware plane has still to be carried.

Data availability

The interbeat-interval records analyzed here are the Normal Sinus Rhythm RR Interval Database [23] and the Fantasia database [24, 25], both openly available from PhysioNet [26]. The scripts that reproduce the exact identities, the discretization laws, the calibrated null, the ground-truth coordinates, the RR cohort and every figure, together with instructions to retrieve the two databases, are available at `https://github.com/sierraporta/ties-complexity-plane`.

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A Proofs of the discretization laws

Throughout, the source is a continuous density $f = F'$ quantized to a lattice of step Δ with bin probabilities $q_k = F((k+\frac{1}{2})\Delta) - F((k-\frac{1}{2})\Delta)$, the alphabet is the weak orders of m elements with $M = a(m)$, and the controlling quantity is the per-pair coincidence probability $c = \sum_k q_k^2$ of (5). We take $f \in L^2 \cap L^\infty$ so that $c = \Delta \int f^2 + O(\Delta^2)$ and the remainders below are uniform in the pattern.

Lemma 1. *For m independent values from the quantized source with coincidence rate c: (i) each of the $m!$ strict orders has probability $\frac{1}{m!}\big(1 - \binom{m}{2}c\big) + O(c^2)$; (ii) each one-tie weak order has probability $\frac{c}{(m-1)!} + O(c^2)$, and there are $\binom{m}{2}(m-1)!$ of them; (iii) patterns with two or more coincidences have total probability $O(c^2)$.*

Proof. A given unordered pair of the m positions lands in the same bin with probability $\sum_k q_k^2 = c$, independently of the order of the remaining values. The $\binom{m}{2}$ pairwise coincidence events are disjoint up to $O(c^2)$, so

the probability of at least one tie is $\binom{m}{2}c+O(c^2)$ and that of all values distinct is $1-\binom{m}{2}c+O(c^2)$. Conditional on all values distinct, exchangeability of the independent sample makes the $m!$ orders equiprobable, which gives (i). For (ii), a specific one-tie pattern fixes the tied pair (probability c) and a strict arrangement of the $m-1$ resulting groups; conditional on exactly one tie the $(m-1)!$ such arrangements are equiprobable, giving $c/(m-1)!$. Two coincidences cost $O(c^2)$, which is (iii). The bin probabilities $q_k \in [0,1]$ and the pattern counts are bounded by M, so the multi-coincidence mass is bounded by $\binom{m}{2}^2 c^2$ independently of the source, making the remainders uniform. □

Proposition 1. $D_0 - D(\Delta) = \dfrac{c}{(m-2)!} + O(c^2)$.

Proof. Write $D = \sum_\pi p_\pi^2 - 1/M$. By Lemma 1 the strict orders contribute $m!\big((1-\binom{m}{2}c)/m!\big)^2 = 1/m! - 2\binom{m}{2}c/m! + O(c^2)$, the one-tie patterns contribute $\binom{m}{2}(m-1)!\,(c/(m-1)!)^2 = O(c^2)$, and higher ties are $O(c^2)$. Hence $D(\Delta) = D_0 - 2\binom{m}{2}c/m! + O(c^2)$, and $2\binom{m}{2}/m! = m(m-1)/m! = 1/(m-2)!$. □

Proposition 2. $\big(H(\Delta)-H_0\big)\log M = \binom{m}{2}\Big[\,c\log\tfrac{1}{c}+(1-\log m)\,c\,\Big]+O(c^2)$, *with leading term* $\binom{m}{2}c\log(1/c)$ *and* $\big(H-H_0\big)\log M/\big[\binom{m}{2}c\log(1/c)\big] = 1+(1-\log m)/\log(1/c)+O(c/\log(1/c))$.

Proof. With $S = -\sum_\pi p_\pi \log p_\pi$ and $p_s = (1-\binom{m}{2}c)/m!$ for a strict order, Lemma 1 and $\log(1-x) = -x + O(x^2)$ give $-m!\,p_s \log p_s = \log(m!) + \binom{m}{2}c\,(1-\log m!) + O(c^2)$. The one-tie patterns, each $p_t = c/(m-1)!$, give $-\binom{m}{2}(m-1)!\,p_t \log p_t = \binom{m}{2}c\log(1/c) + \binom{m}{2}c\log((m-1)!) + O(c^2\log\tfrac{1}{c})$. Adding, with $\log((m-1)!) - \log(m!) = -\log m$, $S(\Delta) - S_0 = \binom{m}{2}c\log(1/c) + \binom{m}{2}c\,(1-\log m) + O(c^2\log\tfrac{1}{c})$, where $S_0 = \log(m!)$. Since $H = S/\log M$ and $H_0 = S_0/\log M$, we have $(H-H_0)\log M = S - S_0$, which is the claim; the ratio follows from $c(1-\log m) = c\log(1/c)\cdot(1-\log m)/\log(1/c)$. □

Proposition 1 read backwards is the estimator (7); its $O(c^2)$ bias is the residual the linear correction leaves. All three results are checked against the exact tie-alphabet coordinates for Gaussian, $m = 3, 4, 5$: the ratio in Proposition 1 tends to one and the two-term form of Proposition 2 tracks the exact entropy shift far better than its leading term alone.

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